

Electricity & Magnetism Homework 2

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Question 1

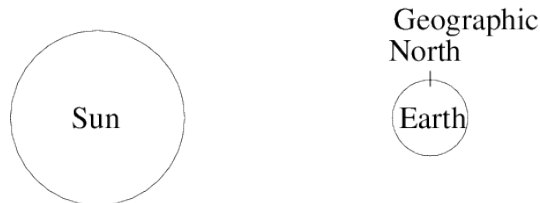
- a) What is the source of the magnetic field?
- b) Why do magnetic field lines only form closed loops?
- c) Draw the magnetic field of a bar magnet. Where is the field strongest? Where is the field weakest?

Question 2

A wire is fixed horizontally and has a current of 100 A running through from left to right. A second wire is placed parallel to it and 10 cm lower. The second wire weighs 100 g/m. How much current needs to be flowing through the second wire and in which direction in order to keep it afloat?

Question 3

A proton, emitted from the sun, is approaching Earth at 1×10^6 m/s perpendicular to the Earth's magnetic field (0.05 mT). The sun is to the left of the Earth as shown in the figure below.



- a) Describe the orbit of the proton. In which direction does it perform circular movement?
- b) Calculate the radius of the proton's orbit.